

1. SUMMARY

This repository is the main codebase for jQuery, a widely used JavaScript library that simplifies DOM manipulation, event handling, animations, and AJAX. The project is currently centered on the 4.x line (main branch), with 3.x in critical-fix-only mode and 1.x/2.x no longer supported. The repo includes build tooling (e.g., `runRollup``), testing utilities (`dataTests``, `testAppend``), CSS/DOM utilities (`curCSS``, `adjustCSS``, `isArrayLike``, `isWindow``), and general core functions (e.g., `jQueryFactory``, `dir``, `q``).

2. KEY FEATURES

- DOM Manipulation – Simplified APIs for selecting elements, traversing the DOM, and modifying content/attributes.
- Event Handling – Cross-browser event binding and delegation (e.g., `delegatedSubmit``, `focusMappedHandler``).
- CSS & Animation Utilities – Functions like `curCSS``, `adjustCSS``, `createFxNow`` for reading/computing styles and driving animations.
- AJAX & Utilities – Core helper utilities such as `isArrayLike``, `isWindow``, and others that support higher-level APIs.
- Build & Release Tooling – Scripts like `runRollup``, `schedule``, and author-related helpers (`compareAuthors``, `sortAuthors``, `checkAuthors``, `getLastModifiedDate``) to support packaging, releases, and governance.
- Testing Infrastructure – Test helpers such as `dataTests``, `testAppend``, `getSelectedOptions`` to validate DOM behaviors and library correctness.

3. TECHNOLOGIES USED

- Language: JavaScript (core library and tooling)
- Build Tooling: Rollup (via `runRollup``) and related Node.js-based scripts
- Ecosystem/Infrastructure:
 - Matrix.org for project meetings
 - Custom testing framework and scripts for browser/DOM validation
- Target Platform: Browser environments (with cross-browser compatibility as a primary concern)